Complete Calabi–Yau Metrics on a Class of Non-compact Abelian-fibered Threefolds

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Problem

For a non-compact Kähler manifold X with trivial canonical bundle, is there a ***complete*** *Calabi-Yau metric on X? If so, what is the behavior of the metric* ***near infinity****?*

Tian–Yau Metrics

The first general result is due to Tian and Yau in their paper [2].

Theorem (Tian–Yau)

Let $D \in |-K_X|$ be a smooth irreducible anti-canonical divisor of a Fano manifold X.

- *The non-compact manifold $M = X \setminus D$ admits a complete Calabi–Yau metric ω_{CY};*
- *The* ***tangent cone at infinity*** *of (M, ω_{CY}) is $\mathbb{R}^+$;*
- *The* ***volume growth of the geodesic ball*** *is $B(x_0, r) \sim O(r^{\frac{2n}{n+1}})$, where $n = \dim M$.*
- *the* ***sectional curvature*** *has decay*

$$|R_m| = \begin{cases} O(r^2) & \text{if } n = 2 \\ O(r^{-\frac{n+1}{n}}) & \text{if } n > 2 \end{cases}$$

Hein's Construction

In 2010, Hans-Joachim Hein [1] constructed complete Calabi–Yau metrics on rational elliptic surfaces with a (singular) fiber removed.

Theorem (Hein)

Let $f : X \to \mathbb{P}^1$ be a rational elliptic surface. For any $p \in \mathbb{P}^1$, $D = f^{-1}(p)$ is an anti-canonical divisor. Set $M = X \setminus D$, and let Ω be a rational holomorphic 2-form on X with $\mathrm{div}(\Omega) = D$.
For any Kähler metric ω on M and sufficiently large $\alpha > 0$, there exists a complete Calabi–Yau metric ω_{CY} on M satisfying $\omega_{CY}^2 = \alpha\,\Omega \wedge \bar{\Omega}$ and $[\omega_{CY}] = [\omega]$.

Remark

1. *Note that rational elliptic surfaces are* ***not Fano****. Hence, Tian–Yau's result does not cover Hein's construction.*
2. *The core analysis technique used by Tian-Yau involves establishing Sobolev inequalities on weighted Hölder spaces to solve the complex Monge–Ampère equation. This method was refined and systematized by Hein and is now referred to as the* ***Tian–Yau–Hein package****.*

The Tian–Yau–Hein Package

Theorem (Tian–Yau–Hein Package)

Let (M, ω_0) be a complete non-compact Kähler manifold admitting $C^{3,\alpha}$ quasi-atlas and satisfying the **SOB**(β) *condition for some $\beta > 0$. Suppose $f \in C^{2,\alpha}(M)$ satisfies $|f| \leq Cr^{-\mu}$ on $\{r > 1\}$ for some $\mu > 2$. If $\beta \leq 2$, further assume $\int_M (e^f - 1)\,\omega_0^m = 0$. Then there exist $\bar{\alpha} \in (0, \alpha]$ and a function $u \in C^{4,\bar{\alpha}}(M)$ such that*

$$(\omega_0 + i\partial\bar{\partial}u)^m = e^f \omega_0^m.$$

If $\beta \leq 2$, then additionally $\int_M |\nabla u|^2 \omega_0^m < \infty$. Moreover, if $f \in C^{k,\bar{\alpha}}_{\mathrm{loc}}(M)$ for some $k \geq 3$, then any such solution u belongs to $C^{k+2,\bar{\alpha}}_{\mathrm{loc}}(M)$.

Gravitational Instanton

Hein's constructions provide examples of four-dimensional hyperkähler manifolds known in physics as **gravitational instantons.**

Definition

A gravitational instanton is a non-compact complete non-flat hyperkähler 4-manifold X satisfying $\int_X |\mathrm{Rm}|^2 < \infty$.

The classification of such instantons was completed by Chen, Chen–Sun–Zhang et al.

Theorem (Chen, Chen–Sun–Zhang)

Gravitational instantons with $|\mathrm{Rm}| = O(r^{-2-\varepsilon})$ fall into four classes: **ALE** *(Asymptotically Locally Euclidean),* **ALF** *(Asymptotically Locally Flat),* **ALG***, and* **ALH***.*
Gravitational instantons with $|\mathrm{Rm}| = O(r^{-2})$ belong to two classes: **ALG*** *and* **ALH****.*

Hein's constructions yield gravitational instantons of types **ALG**, **ALH**, **ALG***, and **ALH***.

Theorem (Hein)

(1) *If D is smooth, then ω_{CY} is* **ALH**, *i.e., asymptotic to a torus bundle over a cylinder.*

(2) *If D is singular with finite monodromy, then ω_{CY} is* **ALG**, *i.e., asymptotic to a torus bundle over a cone.*

(3) *If D is of Kodaira type I_b^*, then ω_{CY} is* **ALG***. *In particular, $|B(x_0, r)| \sim r^2$, the tangent cone at infinity is the cone $\mathbb{R}^2/\mathbb{Z}_2$.*

(4) *If D is of Kodaira type I_b, then ω_{CY} is* **ALH***. *In particular, $|B(x_0, r)| \sim r^{4/3}$, the tangent cone at infinity is $\mathbb{R}^+$.*

The metrics we construct can be viewed as higher-dimensional analogs of these four types, as they exhibit similar asymptotic behavior at infinity.

Main Theorem 1

Theorem (Liang–Z.)

Let A be an abelian surface. Consider the action of $\mathbb{Z}/k\mathbb{Z}$ on $\mathbb{P}^1_{[t:s]} \times A$ defined by

$$(t, x) \mapsto (\zeta_k^{-1} t, \tau \cdot x)$$

where $\zeta_k = \exp(2\pi i/k)$ and $\tau \in \mathrm{Aut}(A)$ satisfies $\tau^*\Omega_A = \zeta_k \Omega_A$ for a holomorphic volume form Ω_A on A.
Let X be the crepant resolution of the quotient $(\mathbb{P}^1_{[t:s]} \times A)/\mathbb{Z}_k$ along the fiber $t = 0$, and let $D = f^*(\{\infty\})$ be the fiber over $t = \infty$. Then for sufficiently large $\lambda > 0$, there exists a Kähler metric ω_{CY} satisfying the complex Monge–Ampère equation

$$\omega_{CY}^3 = \lambda \Omega \wedge \bar{\Omega}$$

for some holomorphic volume form Ω.

Main Theorem 2

Theorem (Liang–Z.)

Let $\pi_i : X_i \to \mathbb{P}^1$ $(i = 1, 2)$ be two rational elliptic surfaces. Fix a singular fiber $F_i \subset X_i$ for each i, and assume both F_1 and F_2 project to the same point $s \in \mathbb{P}^1$, while all other singular fibers project to distinct points. Form the fiber product $f : X = X_1 \times_{\mathbb{P}^1} X_2 \to \mathbb{P}^1$. Then X is a singular variety with all singular loci contained in the fiber $D = f^{-1}(s)$. Set $M = X \setminus D$.

(1) *If the monodromies around F_1 and F_2 are both finite, then for certain specific singular types, M admits a Calabi–Yau metric of type* **ALG** *or* **ALH**.

(2) *If both F_i are of type $I^*_{b_i}$, then M admits a generalized* $\mathbf{ALH}^*$*-type Calabi–Yau metric.*

(3) *If F_1 is of type I^*_b and F_2 is of type II^*, III^*, or IV^*, then M admits a generalized* $\mathbf{ALG}^*$*-type Calabi–Yau metric.*

Main Steps

1. Find an appropriate geometric model;
2. Construct an explicit ansatz near infinity;
3. Prove a $\partial\bar{\partial}$-type lemma to extend the ansatz to the whole space by gluing method;
4. Solve the complex Monge–Ampère equation to perturb the ansatz into a Calabi–Yau metric.

Isotrivial Case

Quotient Model

Near the fiber $t = 0$, the action in Main Theorem 1 is locally given by a subgroup of $SL(3, \mathbb{C})$. The quotient singularity $\mathbb{C}^3/G$ admits a crepant resolution X.
Note that $D = f^*(\{\infty\})$ may carry multiplicity, and $-K_X$ is a rational multiple of D. According to Ueno's classification [3, 4], there are 13 possible group actions.

Non-isotrivial Case

Key Observation

In general, the fiber product of two (minimal) rational elliptic surfaces is itself Calabi–Yau.
However, when we take product of a **non-reduced singular fibers**—specifically of Kodaira types $\mathrm{II}^*, \mathrm{III}^*, \mathrm{IV}^*$, or I_b^*—with other singular fibers, the resulting fiber product is **not normal.**
After take the normalization (and perhaps some more birational surgeries on the singular fiber), we obtain a normal variety with negative canonical divisor.

Examples of Fiber Products

1. If $F_1 = \mathrm{III}$, $F_2 = \mathrm{III}^*$, then on the normalized X we have $K_X = -D/4$.
2. If $F_1 = \mathrm{II}^*$, $F_2 = \mathrm{III}^*$, then on the normalized X we have $K_X = -2D/3$.
3. If $F_1 = I_b^*$, $F_2 = \mathrm{IV}^*$, then on the normalized X we have $K_X = -D/3$.
4. If $F_1 = I_b^*$, $F_2 = I_{b'}^*$, then on the normalized X we have $K_X = -D/2$.

Remark

1. *In complex dimension three, the **relatively minimal model** (ensuring that the canonical divisor is proportional to a fiber) may have **singularities**. Even if it is smooth, it **may not be unique**. Thus, one may not expect a clean classification similar to the case of rational elliptic surfaces.*
2. *The **classification of singular fibers** in higher dimensions remains incomplete. Ueno only handled partial cases, and the remaining ones are highly nontrivial.*
3. ***New (topological) obstructions** arise when the dimension of the fiber increases, making the construction of ansatz much trickier.*

Semi-flat Metric Construction

For general torus fibrations over non-compact Riemann surfaces, one can always define well-behaved **semi-flat metrics.** In particular, for fiber products we have:

Proposition

Let $f : X \to S$ be the fiber product of two elliptic surfaces over a Riemann surface S, both without singular fibers, equipped with a holomorphic section σ and a holomorphic volume form.
Let $U \subset S$ be a coordinate neighborhood with holomorphic coordinate z, and fix an isomorphism

$$X|_U \cong (U \times \mathbb{C}_{v_1} \times \mathbb{C}_{v_2}) \,/\, (\mathbb{Z}\tau_1 + \mathbb{Z}\tau_2 + \mathbb{Z}\tau_3 + \mathbb{Z}\tau_4)\,,$$

where $\tau_1, \tau_2, \tau_3, \tau_4 : U \to \mathbb{C}$ are multivalued holomorphic functions such that $(\tau_1, \tau_2, \tau_3, \tau_4)$ is positively oriented, and σ corresponds to the zero section.

Proposition (continued)

Then $\Omega = g\, dz \wedge dv_1 \wedge dv_2$ *for some holomorphic function* $g : U \to \mathbb{C}$*, and the semi-flat Calabi–Yau metric is given by*

$$\begin{aligned}\omega_{\mathrm{sf},\varepsilon} &= 4i|g|^2 \frac{\operatorname{Im}(\bar\tau_1\tau_2)\operatorname{Im}(\bar\tau_3\tau_4)}{\varepsilon^2} dz \wedge d\bar z \\ &+ \frac{i}{2}\frac{\varepsilon}{\operatorname{Im}(\bar\tau_1\tau_2)}(dv_1 - \Gamma^1 dz) \wedge (d\bar v_1 - \bar\Gamma^1 d\bar z) \\ &+ \frac{i}{2}\frac{\varepsilon}{\operatorname{Im}(\bar\tau_3\tau_4)}(dv_2 - \Gamma^2 dz) \wedge (d\bar v_2 - \bar\Gamma^2 d\bar z),\end{aligned}$$

constructed from σ *and* Ω*, with fiber area equal to* ε*. In particular,* $\omega^3_{\mathrm{sf},\varepsilon} = 6i\,\Omega \wedge \bar\Omega$*. Here,*

$$\Gamma^1(z,v) = \frac{1}{\operatorname{Im}(\bar\tau_1\tau_2)}\left(\operatorname{Im}(\bar\tau_1 v_1)\,\tau_2' - \operatorname{Im}(\bar\tau_2 v_1)\,\tau_1'\right);$$

$$\Gamma^2(z,v) = \frac{1}{\operatorname{Im}(\bar\tau_3\tau_4)}\left(\operatorname{Im}(\bar\tau_3 v_2)\,\tau_4' - \operatorname{Im}(\bar\tau_4 v_2)\,\tau_3'\right).$$

Asymptotics at Infinity

This metric satisfies the **SOB**(β) condition with $\beta \leq 2$, along with all other asymptotic requirements needed for the Tian–Yau–Hein package. In concrete examples, the asymptotic behavior is as follows:

Examples

1. $F_1 = \mathrm{III}, F_2 = \mathrm{III}^*$: asymptotic to a torus bundle over a flat cylinder (**ALG**-type).
2. $F_1 = \mathrm{II}^*, F_2 = \mathrm{III}^*$: asymptotic to asymptotic to a torus bundle over a flat cone of angle $7\pi/6$ (**ALH**-type).
3. $F_1 = I_b^*, F_2 = \mathrm{IV}^*$: the tangent cone at infinity is a flat cone of angle $\pi/3$, with volume growth of order r^2 (**ALG***-type).
4. $F_1 = I_b^*, F_2 = I_{b'}^*$: the tangent cone is $\mathbb{R}^+$, with volume growth of order $r^{3/2}$ (**ALH***-type).

Three Difficulties

1. **First difficulty**: We need a $\partial\bar{\partial}$-lemma on $X \setminus F$. Two main obstructions arise: (i) constructing a Kähler metric ω on $X \setminus F$ whose de Rham cohomology class matches that of $\omega_{\mathrm{sf},\varepsilon}$ in $H^2(X|_{\Delta^*}, \mathbb{R})$; and (ii) an analytic obstruction arises from $H^{0,1}(X|_{\Delta^*})$.
2. **Second difficulty**: ω and $\omega_{\mathrm{sf},\varepsilon}$ may differ significantly in their overlap; naive gluing via cutoff functions might destroy **positivity.**
3. **Third difficulty**: Since $\omega_{\mathrm{sf},\varepsilon}$ satisfies **SOB**(β) with $0 < \beta \leqslant 2$, the Tian–Yau–Hein method requires the **integrability condition**:

$$\int_{X \setminus F} \left(\omega_{\mathrm{sf},\varepsilon}^3 - i\Omega \wedge \bar{\Omega}\right) = 0.$$

Addressing the First Difficulty

1. **Isotrivial case**: By analyzing the crepant resolution explicitly, we glue in explicit special local metrics (e.g., generalized Eguchi–Hanson metrics) near the singularities to obtain the background metric ω.
2. **Non-isotrivial case**: On the product $X_1 \times X_2$, take the product of Ricci-flat metrics from each elliptic surface and pull it back to $X \setminus D$ via the natural embedding to serve as ω.

Proposition ($\partial\bar{\partial}$-Lemma)

There exists a holomorphic section $\tilde{\sigma}$ of f over Δ^* and a smooth function $u_1 : X|_{\Delta^*} \to \mathbb{R}$ such that

$$T^*\omega_{\mathrm{sf},\varepsilon}(1) - \omega = i\partial\bar{\partial}u_1,$$

where T denotes translation relative to the initial section σ.

Example 1 (Isolated Singularities)

Consider the $\mathbb{Z}_3$-action on $\mathbb{P}^1_{[t:s]} \times E \times E$ given by

$$(\zeta_3, (t, w_1, w_2)) \mapsto (\zeta_3 t, \zeta_3 w_1, \zeta_3 w_2),$$

where E is the elliptic curve with hexagonal lattice. The quotient has 9 A_2-**type singularities** along the fiber $t = 0$. The crepant resolution introduces nine exceptional divisors, each modeled on $\mathcal{O}_{\mathbb{P}^2}(-3)$. The generalized Eguchi–Hanson metric on $(\mathbb{C}^3 \setminus \{0\})/\mathbb{Z}_3$ is given by the potential

$$f_a(u) = \sqrt[3]{a^3 + u^3} + \frac{a}{3} \sum_{j=0}^{2} \zeta^j \log \left(\sqrt[3]{1 + \frac{u^3}{a^3}} - \zeta^j \right).$$

This metric is extremely close to the Euclidean metric, allowing direct gluing via cutoff functions.

Example 2 (One-dimensional Singular Locus)

Consider the $\mathbb{Z}_4$-action on $\mathbb{P}^1_{[t:s]} \times E_1 \times E_2$:

$$(i, (t, w_1, w_2)) \mapsto (it, iw_1, -w_2),$$

where E_1 has square lattice and E_2 is arbitrary. The fixed locus consists of four elliptic curves $F_1, \ldots, F_4$ (fixed by a $\mathbb{Z}_2$ subgroup) and eight isolated points P_i (fixed by the full group). The points P_i lie on F_1, F_2, which are quotiented to rational curves, while F_3, F_4 are identified to a single elliptic curve. Since the normal bundles of all F_i are trivial, we can iteratively glue in Eguchi–Hanson-type metrics “from outside in” to construct a global metric.

Addressing the Second and Third Difficulties

These two issues must be handled simultaneously.

Strategy

On the overlap region, add a **positive** $(1,1)$**-form** $t\beta$ pulled back from the base (with $t > 0$) to ensure **positivity**.
Introduce a **second parameter** α to satisfy the **integrability condition.**
Specifically, define a family of semi-flat metrics $\omega_{\mathrm{sf},\varepsilon}(\alpha)$ at infinity such that

$$\omega_{\mathrm{sf},\varepsilon}(\alpha) - \omega_{\mathrm{sf},\varepsilon}(1) = (\alpha - 1) i\partial\bar{\partial}u,$$

and $\omega_{\mathrm{sf},\varepsilon}(\alpha)^3 = \alpha\,\Omega \wedge \bar{\Omega}$.

Glued Metric

Choose parameters r, s, C_0 and a positive $(1,1)$-form β on $\mathbb{P}^1$. For $\alpha > 0$ and $t > 0$, define

$$\omega_0(\alpha, t) := \begin{cases} \omega + t\beta + i\partial\bar{\partial}(\psi\tilde{u}_\alpha) & \text{on } M|^{\complement}_{\Delta(r)}; \\ \omega + t\beta + i\partial\bar{\partial}u_\alpha & \text{on } \Delta(r+s)^*, \end{cases}$$

where

$$\tilde{u}_\alpha = u_1 + (\alpha - 1)(u - v) \quad \text{on } \Delta(r+3s) \setminus \Delta(r),$$

with v the harmonic function matching u on the boundary, and $\psi = \psi(r, s)$ a suitable cutoff function. Then $\omega_0(\alpha, t)$ equals ω outside $M|_{\Delta(r+3s)}$ and coincides with $T^*\omega_{\mathrm{sf},\varepsilon}(\alpha)$ on $\Delta(r)^*$.

Define

$$I(\alpha, t) := \int_M \left(\omega_0(\alpha, t)^3 - \alpha\, \Omega \wedge \bar{\Omega}\right).$$

We seek α, t such that $\omega_0(\alpha, t)$ is positive definite and $I(\alpha, t) = 0$.

Key Observation

The crucial observation is that $I(\alpha, t)$ is **bilinear** in α and t. Choose $t = t(\alpha)$ as a linear function of α so that $I(\alpha, t(\alpha))$ becomes a **linear function in α.** Then select parameters so that $I(0, t(0)) > 0$ and $I'(0, t(0)) < 0$, guaranteeing a solution $\alpha > 0$ with $I(\alpha, t(\alpha)) = 0$.

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Thank you!